

Exercises

1. Use homotopy groups to show there is no retraction $\mathbb{R}P^n \rightarrow \mathbb{R}P^k$ if $n > k > 0$.
2. Show the action of $\pi_1(\mathbb{R}P^n)$ on $\pi_n(\mathbb{R}P^n) \approx \mathbb{Z}$ is trivial for n odd and nontrivial for n even.
3. Let X be obtained from a lens space of dimension $2n + 1$ by deleting a point. Compute $\pi_{2n}(X)$ as a module over $\mathbb{Z}[\pi_1(X)]$.
4. Let $X \subset \mathbb{R}^{n+1}$ be the union of the infinite sequence of spheres S_k^n of radius $1/k$ and center $(1/k, 0, \dots, 0)$. Show that $\pi_i(X) = 0$ for $i < n$ and construct a homomorphism from $\pi_n(X)$ onto $\prod_k \pi_n(S_k^n)$.
5. Let $f: S_\alpha^2 \vee S_\beta^2 \rightarrow S_\alpha^2 \vee S_\beta^2$ be the map which is the identity on the S_α^2 summand and which on the S_β^2 summand is the sum of the identity map and a homeomorphism $S_\beta^2 \rightarrow S_\alpha^2$. Let X be the mapping torus of f , the quotient space of $(S_\alpha^2 \vee S_\beta^2) \times I$ under the identifications $(x, 0) \sim (f(x), 1)$. The mapping torus of the restriction of f to S_α^2 forms a subspace $A = S^1 \times S_\alpha^2 \subset X$. Show that the maps $\pi_2(A) \rightarrow \pi_2(X) \rightarrow \pi_2(X, A)$ form a short exact sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$, and compute the action of $\pi_1(A)$ on these three groups. In particular, show the action of $\pi_1(A)$ is trivial on $\pi_2(A)$ and $\pi_2(X, A)$ but is nontrivial on $\pi_2(X)$.
6. Show that the relative form of the Hurewicz theorem in dimension n implies the absolute form in dimension $n - 1$ by considering the pair (CX, X) where CX is the cone on X .
7. Construct a CW complex X with prescribed homotopy groups $\pi_i(X)$ and prescribed actions of $\pi_1(X)$ on the $\pi_i(X)$'s.
8. Show the suspension of an acyclic CW complex is contractible.
9. Show that a map between simply-connected CW complexes is a homotopy equivalence if its mapping cone is contractible. Use the preceding exercise to give an example where this fails in the nonsimply-connected case.
10. Let the CW complex X be obtained from $S^1 \vee S^n$, $n \geq 2$, by attaching a cell e^{n+1} by a map representing the polynomial $p(t) \in \mathbb{Z}[t, t^{-1}] \approx \pi_n(S^1 \vee S^n)$, so $\pi_n(X) \approx \mathbb{Z}[t, t^{-1}]/(p(t))$. Show $\pi'_n(X)$ is cyclic and compute its order in terms of $p(t)$. Give examples showing that the group $\pi_n(X)$ can be finitely generated or not, independently of whether $\pi'_n(X)$ is finite or infinite.
11. Let X be a connected CW complex with 1-skeleton X^1 . Show that $\pi_2(X, X^1) \approx \pi_2(X) \times K$ where K is the kernel of $\pi_1(X^1) \rightarrow \pi_1(X)$, a free group. Show also that $\pi'_2(X, X^1) \approx \pi'_2(X) \times K'$ where K' is the quotient of K obtained by factoring out the conjugation action of $\pi_1(X^1)$. Note that K' is abelian.
12. Show that a map $f: X \rightarrow Y$ of connected CW complexes is a homotopy equivalence if it induces an isomorphism on π_1 and if a lift $\tilde{f}: \tilde{X} \rightarrow \tilde{Y}$ to the universal covers induces an isomorphism on homology. [The latter condition can be restated in terms of

homology with local coefficients as saying that $f_*: H_*(X; \mathbb{Z}[\pi_1 X]) \rightarrow H_*(Y; \mathbb{Z}[\pi_1 Y])$ is an isomorphism; see §3.H.]

13. Show that a map between connected n -dimensional CW complexes is a homotopy equivalence if it induces an isomorphism on π_i for $i \leq n$. [Pass to universal covers and use homology.]

14. If an n -dimensional CW complex X contains a subcomplex Y homotopy equivalent to S^n , show that the map $\pi_n(Y) \rightarrow \pi_n(X)$ induced by inclusion is injective. [Use the Hurewicz homomorphism.]

15. Show that a closed simply-connected 3-manifold is homotopy equivalent to S^3 . [Use Poincaré duality, and also the fact that closed manifolds are homotopy equivalent to CW complexes, from Corollary A.12 in the Appendix. The stronger statement that a closed simply-connected 3-manifold is homeomorphic to S^3 is still unproved. This is the *Poincaré conjecture*, without doubt the most famous open problem in topology. The higher-dimensional analog, that a closed n -manifold homotopy equivalent to S^n is homeomorphic to S^n , has been proved for all $n \geq 4$.]

16. Show that the closed surfaces with infinite fundamental group are $K(\pi, 1)$'s by showing that their universal covers are contractible, via the Hurewicz theorem and results of §3.3.

17. Show that the map $\langle X, Y \rangle \rightarrow \text{Hom}(\pi_n(X), \pi_n(Y))$, $[f] \mapsto f_*$, is a bijection if X is an $(n-1)$ -connected CW complex and Y is a path-connected space with $\pi_i(Y) = 0$ for $i > n$. Deduce that CW complex $K(G, n)$'s are uniquely determined, up to homotopy type, by G and n .

18. If X and Y are simply-connected CW complexes such that $\tilde{H}_i(X)$ and $\tilde{H}_j(Y)$ are finite and of relatively prime orders for all pairs (i, j) , show that the inclusion $X \vee Y \hookrightarrow X \times Y$ is a homotopy equivalence and $X \wedge Y$ is contractible. [Use the Künneth formula.]

19. If X is a $K(G, 1)$ CW complex, show that $\pi_n(X^n)$ is free abelian for $n \geq 2$.

20. Let G be a group and X a simply-connected space. Show that for the product $K(G, 1) \times X$ the action of π_1 on π_n is trivial for all $n > 1$.

21. Given a sequence of CW complexes $K(G_n, n)$, $n = 1, 2, \dots$, let X_n be the CW complex formed by the product of the first n of these $K(G_n, n)$'s. Via the inclusions $X_{n-1} \subset X_n$ coming from regarding X_{n-1} as the subcomplex of X_n with n^{th} coordinate equal to a basepoint 0-cell of $K(G_n, n)$, we can then form the union of all the X_n 's, a CW complex X . Show $\pi_n(X) \approx G_n$ for all n .

22. Show that $H_{n+1}(K(G, n); \mathbb{Z}) = 0$ if $n > 1$. [Build a $K(G, n)$ from a Moore space $M(G, n)$ by attaching cells of dimension $> n + 1$.]

23. Extend the Hurewicz theorem by showing that if X is an $(n-1)$ -connected CW complex, then the Hurewicz homomorphism $h: \pi_{n+1}(X) \rightarrow H_{n+1}(X)$ is surjective

when $n > 1$, and when $n = 1$ show there is an isomorphism $H_2(X)/h(\pi_2(X)) \approx H_2(K(\pi_1(X), 1))$. [Build a $K(\pi_n(X), n)$ from X by attaching cells of dimension $n + 2$ and greater, and then consider the homology sequence of the pair (Y, X) where Y is X with the $(n + 2)$ -cells of $K(\pi_n(X), n)$ attached. Note that the image of the boundary map $H_{n+2}(Y, X) \rightarrow H_n(X)$ coincides with the image of h , and $H_{n+1}(Y) \approx H_{n+1}(K(\pi_n(X), n))$. The previous exercise is needed for the case $n > 1$.]

24. Show there is a Moore space $M(G, 1)$ with $\pi_1(M(G, 1)) \approx G$ iff $H_2(K(G, 1); \mathbb{Z}) = 0$. [Use the preceding problem. Build such an $M(G, 1)$ from the 2-skeleton K^2 of a $K(G, 1)$ by attaching 3-cells according to a basis for the free group $H_2(K^2; \mathbb{Z})$.] In particular, there is no $M(\mathbb{Z}^n, 1)$ with fundamental group $\mathbb{Z}^n$, free abelian of rank n , if $n \geq 2$.

25. Let X be a CW complex with $\pi_i(X) = 0$ for $1 < i < n$ for some $n \geq 2$. Show that $H_n(X)/h(\pi_n(X)) \approx H_n(K(\pi_1(X), 1))$, where h is the Hurewicz homomorphism.

26. Generalizing the example of $\mathbb{R}P^2$ and $S^2 \times \mathbb{R}P^\infty$, show that if X is a connected finite-dimensional CW complex with universal cover $\tilde{X}$, then X and $\tilde{X} \times K(\pi_1(X), 1)$ have isomorphic homotopy groups but are not homotopy equivalent if $\pi_1(X)$ contains elements of finite order.

27. From Lemma 4.39 deduce that the image of the map $\pi_2(X, x_0) \rightarrow \pi_2(X, A, x_0)$ lies in the center of $\pi_2(X, A, x_0)$.

28. Show that the group $\mathbb{Z}_p \times \mathbb{Z}_p$ with p prime cannot act freely on any sphere S^n , by filling in details of the following argument. Such an action would define a covering space $S^n \rightarrow M$ with M a closed manifold. When $n > 1$, build a $K(\mathbb{Z}_p \times \mathbb{Z}_p, 1)$ from M by attaching a single $(n + 1)$ -cell and then cells of higher dimension. Deduce that $H^{n+1}(K(\mathbb{Z}_p \times \mathbb{Z}_p, 1); \mathbb{Z}_p)$ is $\mathbb{Z}_p$ or 0, a contradiction. (The case $n = 1$ is more elementary.)

29. Finish the homotopy classification of lens spaces begun in Exercise 2 of §3.E by showing that two lens spaces $L_m(\ell_1, \dots, \ell_n)$ and $L_m(\ell'_1, \dots, \ell'_n)$ are homotopy equivalent if $\ell_1 \cdots \ell_n \equiv \pm k^n \ell'_1 \cdots \ell'_n \pmod{m}$ for some integer k , via the following steps:

- Reduce to the case $k = 1$ by showing that $L_m(\ell'_1, \dots, \ell'_n) = L_m(k\ell'_1, \dots, k\ell'_n)$ if k is relatively prime to m . [Rechoose the generator of the $\mathbb{Z}_m$ action on S^{2n-1} .]
- Let $f: L \rightarrow L'$ be a map constructed as in part (b) of the exercise in §3.E. Construct a map $g: L \rightarrow L'$ as a composition $L \rightarrow L \vee S^{2n-1} \rightarrow L \vee S^{2n-1} \rightarrow L'$ where the first map collapses the boundary of a small ball to a point, the second map is the wedge of the identity on L and a map of some degree d on S^{2n-1} , and the third map is f on L and the projection $S^{2n-1} \rightarrow L'$ on S^{2n-1} . Show that g has degree $k_1 \cdots k_n + dm$, that is, g induces multiplication by $k_1 \cdots k_n + dm$ on $H_{2n-1}(-; \mathbb{Z})$. [Show first that a lift of g to the universal cover S^{2n-1} has this degree.]

- (c) If $\ell_1 \cdots \ell_n \equiv \pm \ell'_1 \cdots \ell'_n \pmod{m}$, choose d so that $k_1 \cdots k_n + dm = \pm 1$ and show this implies that g induces an isomorphism on all homotopy groups, hence is a homotopy equivalence. [For π_i with $i > 1$, consider a lift of g to the universal cover.]

30. Let E be a subspace of $\mathbb{R}^2$ obtained by deleting a subspace of $\{0\} \times \mathbb{R}$. For which such spaces E is the projection $E \rightarrow \mathbb{R}$, $(x, y) \mapsto x$, a fiber bundle?

31. For a fiber bundle $F \rightarrow E \rightarrow B$ such that the inclusion $F \hookrightarrow E$ is homotopic to a constant map, show that the long exact sequence of homotopy groups breaks up into split short exact sequences giving isomorphisms $\pi_n(B) \approx \pi_n(E) \oplus \pi_{n-1}(F)$. In particular, for the Hopf bundles $S^3 \rightarrow S^7 \rightarrow S^4$ and $S^7 \rightarrow S^{15} \rightarrow S^8$ this yields isomorphisms

$$\begin{aligned}\pi_n(S^4) &\approx \pi_n(S^7) \oplus \pi_{n-1}(S^3) \\ \pi_n(S^8) &\approx \pi_n(S^{15}) \oplus \pi_{n-1}(S^7)\end{aligned}$$

Thus $\pi_7(S^4)$ and $\pi_{15}(S^8)$ contain $\mathbb{Z}$ summands.

32. Show that if $S^k \rightarrow S^m \rightarrow S^n$ is a fiber bundle, then $k = n - 1$ and $m = 2n - 1$. [Look at the long exact sequence of homotopy groups.]

33. Show that if there were fiber bundles $S^{n-1} \rightarrow S^{2n-1} \rightarrow S^n$ for all n , then the groups $\pi_i(S^n)$ would be finitely generated free abelian groups computable by induction, and nonzero for $i \geq n \geq 2$.

34. Let $p: S^3 \rightarrow S^2$ be the Hopf bundle and let $q: T^3 \rightarrow S^3$ be the quotient map collapsing the complement of a ball in the 3-dimensional torus $T^3 = S^1 \times S^1 \times S^1$ to a point. Show that $pq: T^3 \rightarrow S^2$ induces the trivial map on π_* and $\tilde{H}_*$, but is not homotopic to a constant map.

35. Show that the fiber bundle $S^3 \rightarrow S^{4n+3} \rightarrow \mathbb{H}P^n$ gives rise to a quotient fiber bundle $S^2 \rightarrow \mathbb{C}P^{2n+1} \rightarrow \mathbb{H}P^n$ by factoring out the action of S^1 on S^{4n+3} by complex scalar multiplication.

36. For basepoint-preserving maps $f: S^1 \rightarrow X$ and $g: S^n \rightarrow X$ with $n > 1$, show that the Whitehead product $[f, g]$ is $\pm(g - fg)$, where fg denotes the action of f on g .

37. Show that all Whitehead products in a path-connected H-space are trivial.

38. Show $\pi_3(S^1 \vee S^2)$ is not finitely generated as a module over $\mathbb{Z}[\pi_1(S^1 \vee S^2)]$ by considering Whitehead products in the universal cover, using the results in Example 4.52. Generalize this to $\pi_{i+j-1}(S^1 \vee S^i \vee S^j)$ for $i, j > 1$.

39. Show that the indeterminacy of a Toda bracket $\langle f, g, h \rangle$ with $f \in \pi_i^S$, $g \in \pi_j^S$, $h \in \pi_k^S$ is the subgroup $f \cdot \pi_{j+k+1}^S + h \cdot \pi_{i+j+1}^S$ of $\pi_{i+j+k+1}^S$.

4.3 Connections with Cohomology

The Hurewicz theorem provides a strong link between homotopy groups and homology, and hence also an indirect relation with cohomology. But there is a more direct connection with cohomology of a quite different sort. We will show that for every CW complex X there is a natural bijection between $H^n(X; G)$ and the set $\langle X, K(G, n) \rangle$ of basepoint-preserving homotopy classes of maps from X to a $K(G, n)$. We will also define a natural group structure on $\langle X, K(G, n) \rangle$ that makes the bijection a group isomorphism. The mere fact that there is any connection at all between cohomology and homotopy classes of maps is the first surprise here, and the second is that Eilenberg–MacLane spaces are involved, since their definition is entirely in terms of homotopy groups, which on the face of it have nothing to do with cohomology.

After proving this basic isomorphism $H^n(X; G) \approx \langle X, K(G, n) \rangle$ and describing a few of its immediate applications, the later parts of this section aim toward a further study of Postnikov towers, which were introduced briefly in §4.1. These provide a general theoretical method for realizing an arbitrary CW complex as a sort of twisted product of Eilenberg–MacLane spaces, up to homotopy equivalence. The most geometric interpretation of the phrase ‘twisted product’ is the notion of fiber bundle introduced in the previous section, but here we need the more homotopy-theoretic notion of a *fibration*, so before we begin the discussion of Postnikov towers we first take a few pages to present some basic constructions and results about fibrations.

As we shall see, Postnikov towers can be expressed as sequences of fibrations with fibers Eilenberg–MacLane spaces, so we can again expect close connections with cohomology. One such connection is provided by *k-invariants*, which describe, at least in principle, how Postnikov towers for a broad class of spaces are determined by a sequence of cohomology classes. Another application of these ideas, described at the end of the section, is a technique for factoring basic extension and lifting problems in homotopy theory into a sequence of smaller problems whose solutions are equivalent to the vanishing of certain cohomology classes. This technique goes under the somewhat grandiose title of Obstruction Theory, though it is really quite a simple idea when expressed in terms of Postnikov towers.

The Homotopy Construction of Cohomology

The main result of this subsection is the following fundamental relationship between singular cohomology and Eilenberg–MacLane spaces:

Theorem 4.57. *There are natural bijections $T : \langle X, K(G, n) \rangle \rightarrow H^n(X; G)$ for all CW complexes X and all $n > 0$, with G any abelian group. Such a T has the form $T([f]) = f^*(\alpha)$ for a certain distinguished class $\alpha \in H^n(K(G, n); G)$.*